Problems in ML4H

Day 1 — What we're actually trying to solve

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Why this week exists

Every health AI project lives or dies in one gap:

an interesting idea → a system that actually helps patients.

- A strong notebook number is not clinical impact.
- The distance between the two is where projects quietly die.
- This week walks that gap end to end — **each day closes one segment.**

The seven gaps — the week on one slide

DAY	FROM…	…TO
1	an interesting topic	a well-posed **problem**
2	a problem	the **data** to attack it
3	data	a credible **evaluation**
4	a target	a **method** that reaches it
5	a model	a **deployed** system
6	a prototype	a **resourced** program
7	seven deliverables	one **synthesis**

FIG F1.1 **The seven-gap arc.** Horizontal "spine" diagram: seven chevrons, idea → bedside, each labeled with its day. This is the master diagram of the week — reused on every later day's handoff slide. *See build guide F1.1.*

Today is **FRAME**: an interesting topic → a real, well-posed problem.

How today runs

- **Now (9:15–10:30):** this lecture — the conceptual frame.
- **Then:** small group — you state your problem aloud.
- **1:30:** guest — Emily Alsentzer takes ONE real project through the messy reality of problem formulation.
- **By end of day:** Workbook Part 1 — your problem statement.

The lecture is abstract on purpose; the day makes it concrete.

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THE FIRST QUESTION

What *is* ML4H?

Is health AI just "machine learning, applied to hospitals"?

The central claim

ML4H is **not** a domain application of ML — it is a distinct field.

The steelman against this — *"isn't it just sequence/tabular ML with domain quirks?"* — fails on two structural points:

- The **label is endogenous** to the system you predict: it is produced by the same care process that generates the features.
- **Deployment changes the data-generating process** — prediction in health is *performative*, not passive (Perdomo et al., 2020).

These aren't quirks: they **complicate identifiability** for causal and decision claims, and **violate the i.i.d./stationarity** assumptions predictive workflows lean on.

Forty years of perspective

What is **genuinely new** now — and what is **recycled** under a new name.

From **MYCIN** and **INTERNIST** to today's transformers, the cycle repeats: each era promised autonomy and delivered, at its best, useful **decision support**.

FIG F1.2 **An ML4H timeline.** A horizontal timeline from expert systems (1970s–80s) → statistical ML on clinical data (1990s–2000s) → deep learning on EHR/imaging (2010s) → foundation models (2020s). Annotate each era with "the promise" and "what actually stuck." *See build guide F1.2.*

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THE COMMON STRUCTURE

What every health problem shares

Six features hold across *all* health problems

Regardless of domain or modality:

- **High stakes** — decisions affect lives.
- **Distribution shift** — patients, practice, coding all change.
- **Label noise** — "ground truth" is a clinical judgment.
- **Causal structure** — treatment and outcome entangle.
- **Regulation** — health AI is governed, not free.
- **A human in the loop** — a clinician mediates the decision.

FIG F1.3 **The six-feature wheel.** A hub-and-spoke graphic: "Health problem" at center, six labeled spokes. Reused as a recurring motif — later days light up the spoke they address. *See build guide F1.3.*

High stakes → asymmetric, patient-specific cost

In health, the two error types are **not** interchangeable.

- A missed sepsis case and a false sepsis alarm cost wildly different things.
- The cost is **patient-specific** — and it is rarely the loss your model optimizes.
- "Accuracy" averages over a cost structure that is anything but uniform.

> The metric you train on is a stand-in for a clinical cost you have not written down. → Day 3: estimands

Distribution shift → the data moves under you

The IID assumption is a convenience, not a fact. Two points past the textbook:

- Shift has a **causal direction.** Predicting effect-from-cause (anticausal) degrades differently than cause-from-effect; what is invariant across sites depends on the causal graph, not the marginal (Schölkopf et al., 2012).
- **More sites ≠ more general.** Transportability is a property of *which* mechanism is stable; naively pooling biased sources can *compound* bias rather than average it out.

Concretely: an ICD-9 → ICD-10 cutover moves label prevalence with no change in biology — the model learns the **calendar**, not the disease.

→ Day 2: data fusion

→ Day 5: transport & monitoring

Label noise → "ground truth" is a judgment

The label is usually a **proxy**, recorded for another purpose — so the open problem isn't *handling* noise, it's *constructing* labels.

- The outcome you can measure ≠ the outcome you care about; the noise is often **systematic and label-dependent**, not symmetric.
- The live methodology: **programmatic weak supervision** and **anchor learning** — build labels from noisy sources you can reason about, rather than trusting one column.
- Inter-rater disagreement isn't noise to average away; it can encode the **subgroup** the model will later fail.

→ Day 2: the label is a byproduct of care

Causal structure → treatment and outcome entangle

Most clinical data is generated under **treatment** — so the question is *identifiability*, not just confounding.

- Confounding by indication is the symptom; the disease is that the effect you want is **not identified** from the observational distribution without assumptions.
- Naming those assumptions explicitly — and emulating the **target trial** you can't run — is the discipline (Hernán & Robins).
- A predictor fit on treated data can be **anti-useful** for deciding *whether to treat*: it has learned the current policy.

This is why some health questions are **not** prediction problems at all.

Why health breaks generic ML assumptions

The intellectual core of the lecture:

- **IID is rarely true** — see distribution shift.
- **The label is often a proxy** — see label noise.
- **"Performance" ≠ "benefit"** — a better AUROC need not help a patient.
- **Error cost is asymmetric and patient-specific** — see high stakes.

Generic ML assumes these away. **Health does not let you.**

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PROMISE AND LIMITS

What AI can — and cannot — do in health

Where AI is genuinely well-suited

The honest optimistic half:

- **Pattern recognition at scale** — especially high-dimensional data.
- **Triage and prioritization** — directing scarce human attention.
- **Surfacing signal** a human reader would miss or lack time to find.
- **Tasks with abundant, well-defined labels.**

Worked example — mammography risk modeling

A real, credible contribution (Yala et al., 2021):

- A model that estimates **future** breast-cancer risk from a mammogram.
- Beats traditional risk factors; works across populations when built carefully.
- Pattern recognition at scale, on a task with **clear labels** and a **real decision** (screening interval).

FIG F1.4 **Risk-model schematic.** Mammogram → model → risk score → screening decision. Pull a figure from Yala et al. 2021 (cite) or generate a clean schematic. *See build guide F1.4.*

Where AI is poorly suited — or cannot help

- Problems that are **actually causal**, not predictive.
- Settings with **no credible label** to learn from.
- Problems where the binding constraint is **not prediction** but action, access, or trust.

> The most common mistake is reaching for a predictor when the decision needs a causal effect — or needs a policy change, not a model.

AI does not establish causation from observational data alone, does not transfer across settings without evidence, and does not absorb clinical accountability. A person remains responsible.

Prediction, causation, decision

Three different questions. Most projects conflate them.

Prediction
What will happen?
Forecast the outcome.

Causation
What if we act?
Estimate an effect.

Decision
What should we do?
Choose under a loss.

> Kleinberg et al. (2015) — the *prediction policy problem*: some decisions need an accurate forecast; others need the effect of acting.

The harder point: the boundary **moves once you deploy.** A forecast that triggers action becomes an intervention — so even "pure prediction" projects must write down an **estimand** (ICH E9(R1)), not just a metric.

→ Day 3: estimands

Worked contrast — readmission

The same clinical situation, two different questions:

- **Prediction:** "Which patients will be readmitted within 30 days?"
  → a forecast; useful for triage and resource planning.
- **Causation:** "Does *this intervention* reduce readmission?"
  → an effect; a **target-trial** question (population, treatment, contrast, outcome, time-zero).

A perfect readmission **predictor** tells you nothing about whether your **program** works — and the readmission penalty era shows the policy gap is not hypothetical.

THE SIGNATURE MOVE

A problem is not a method

Start from the need, not the technique

For this room the trap is subtler than "I want to use <method>":

- Much of ML4H optimizes **benchmarks that lack construct validity** — a leaderboard is not evidence the underlying problem is real or well-posed (Raji et al., 2021).
- "State-of-the-art on <dataset>" can be a method in search of a problem, dressed as progress.

> **The test:** can you state your problem *without naming a method or a benchmark?*
> If you can't, you have a technique — or a leaderboard — in search of a problem.

Problem formulation is itself a choice

One clinical situation can be posed as **many** ML problems — and the framing has consequences **before any model exists.**

- The choice of target, population, and label encodes values (Passi & Barocas, 2019).
- **Fairness and harm are baked in at formulation** — not added later.
- A method cannot rescue a badly chosen problem.

FIG F1.5 **One situation, many problems.** A fan diagram: a single clinical situation (center) branching to 3–4 distinct ML problem formulations, each with a different target/label and a different downstream harm. *See build guide F1.5.*

Worked case — "reduce healthcare cost"

The same goal, two formulations, two very different systems:

Predict cost

Target = future spending.

Optimizes for who is expensive.

Predict need

Target = unmet clinical need.

Optimizes for who is sick.

Cost is a **proxy** for need — and the proxy is biased by who has had access to care. Same data, opposite consequences.

Obermeyer et al. 2019 — returns on Day 5

Stakeholders and the unmet need

A well-posed problem **names the people.**

- **Patients** — bear the outcome.
- **Clinicians** — act on or override the output.
- **Health systems** — absorb cost and workflow change.
- **Payers** — decide what is reimbursed.
- **Regulators** — decide what is permitted.
- **The public** — bears externalities, grants trust.

FIG F1.6 **Stakeholder map.** The patient at the center, five surrounding stakeholders, each annotated with "their definition of success." *See build guide F1.6.*

Each has different incentives — and a **different definition of "success."**

New vs. recycled — a senior, candid read

What in *this* moment is genuinely new — and what is recurring hype?

Plausibly new

- Scale of routinely-collected EHR data
- Modern representation learning
- Foundation models as reusable substrate

Recycled / overclaimed

- "AI will replace clinicians" — said of expert systems in the 1980s
- Benchmarks mistaken for deployment
- New names for old shortcuts

> The failure modes recur too: misspecification, uninterpretability, and irresponsibility are old fallacies in new clothes. Design principles & fallacies — Szolovits et al. 2023

Open problems — what we genuinely don't know

- **Identification at scale** — when can an effect be trusted from observational health data without a trial?
- **Foundation models for health** — transferable clinical structure, or laundered dataset shortcuts? → Day 4
- **Evaluation that predicts deployment value** — our benchmarks rarely do. → Day 3
- **Generative & agentic clinical systems** — no ground truth, a new failure surface.

> The field's hardest questions are open. Your project can move one of them.

What makes a *good* problem

The rubric for today's deliverable:

1. A **clear unmet need** — not a technique looking for a use.
2. **Identifiable stakeholders** and an explicit definition of success.
3. A **plausible path** — real and pursuable within ~a year.
4. **The test:** can you state it *without naming a method?*

Your week, your Workbook

Your **project** is the real assignment — our lectures serve it.

> **Today's deliverable — Problem statement (1 paragraph).**
> Name the **unmet need**, the **stakeholders**, and the **structural gap** your project addresses. State it without naming a method.

Next: 11:00 small group — state your problem aloud · 1:30 guest (Emily Alsentzer) — one real project lived through, from messy beginnings.